

Enhancing agricultural research with an Attention-Based Hybrid Model for precise classification of rice varieties

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ABSTRACT

As a staple food feeding over half of the world's population, rice needs well defined classification techniques to improve agricultural yields, the supply chain, and food safety. To meet these needs, this work presents an Attention-Based Hybrid Model that can accurately and effectively classify Bangladeshi rice varieties. This research covers complex variations of feature space in shape, texture, and color based on 20 different rice varieties by using a comprehensive dataset of 27,000 high-resolution images that include real-world agricultural issues. The core innovation is proposed based on the Attention-Based CNN and CBAM structure. This in fact effectively highlights and enhances spatially and channel-orientated features, allowing the model to tell apart morphologically similar types of rice with high accuracy. The proposed Attention-Based CNN had achieved an accuracy of 91.92%, which leads to an improvement in both generalization aspects and robustness across different categories of testing conditions. Moreover, extending the proposed framework, feature extraction combined with KNN classifier reported the top accuracy of 99.35% proving that modern feature extraction and classification algorithms go hand in hand. This new combined approach does better than Random Forest and Support Vector Classifier because it solves problems that are normally associated with it such as finegrained features and scaling. Apart from that, the model represents one level up from the current paradigm of automated agriculture, bringing a robust, standardized, and flexible solution for rice variety identification. That way, this study provides a way of connecting a technological standpoint with the requirements that farmers have to address in order to further the issue of sustainability, support precision agriculture, and address the growing need for food quality and security in the world.

1. Introduction

As a vital food crop, rice is essential to feeding almost half of the world's population. Due to changing dietary habits and population expansion, there will likely be a significant increase in demand for rice in the upcoming years (Abhishek, Anjali, & Raj, 2023). In order to meet the varied demands of customers and global markets, precise and effective rice categorization techniques are essential. These techniques allow for greater quality control, enhanced marketability, and improved processing.

Rice is cereal grain that is produced in many countries, and is eaten in all parts of the globe. Among of the factors that determine its market value are texture, shape, color, and fracture rate (Aukkapinyo, Sawangwong, Pooyoi, & Kusakunniran, 2020). It is, therefore, necessary to evaluate the surface characteristics of pre-processed rice grains in order to establish marketability, storage stability, processing quality, and consumer satisfaction. However, measuring quality and characteristics

of the surface is often time-consuming, random, as well as approximate with regular methodologies (Moses et al., 2022). To overcome these problems, recently developed advanced computerized techniques are being applied to analyze rice grain classification faster, accurately and effectively.

Bangladesh is currently the world's third-largest producer of rice (Mamun et al., 2021). As the people increase across the globe a demand of 14,886 million ton (MT) food is expected to be needed in 2050 (Islam & Karim, 2019). Because the grain evaluation and classification through manual method largely relies on human skill, they are rather expensive and time consuming. As a result of this disparity in the profession experience of the specialists involved, the evaluation process in manual approaches often varies. Even if the quality requirements have to be standardized due to international character of trading and distribution, it is difficult to set them because of these discrepancies in manual processes. In addition, when the type

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of assessments is large, it is hard to make decisions faster and be consistent while making them (Koklu, Cinar, & Taspinar, 2021). These limitations could possibly be addressed through automated approaches to HTR and stellar candidates can then be characterized with high-throughput methods with minimum human interference as well as ensuring consistent results across large datasets. Moreover, while there may be a physical distinction between paddy variety and variety or sub variety, one may find that two apparently similar varieties may not genetically be similar (Kadam, Pabrekar, Sawardekar, & Barage, 2023). This in turn raises the total market value of rice products and underscores the need of developing dependable, technologically advanced techniques that can distinguish and classify varying types of rice as is required for this method.

The typical methods for rice variety classification, based on state-of-the-art approaches, show moderate efficiency, and numerous drawbacks that prevent the achievement of more effective results from classifications. As a result of reducing dimensionality, conventional feature extraction techniques, Principal Component Analysis (PCA), and Linear Discriminant Analysis (LDA), rely on linear transformations (Fabiya et al., 2020; Weng et al., 2020). All these techniques fail to provide proper solution for rhythmic patterns and the subtle, non-linear patterning characteristics involved in high random data such as texture, contour and minor slight variations in the rice grains. This leads to poor classification since they tend to overlook relevant data most of the time.

Second, due to reliance on preexisting features and domain expertise, application of traditional, manual, or hand-designed, feature extraction techniques such as HOG or GLCM cannot be as thoroughly used (Ahmed, Ali, & Khan, 2021; Setiawan, Adi, & Widodo, 2023). Since all these methods apply uniform operation to all input data, they are incapable of handling changes in the illumination, perspective, or surface roughness in interactions that define visual perception of rice grains. For this reason, they are unable to easily identify which features are most differentiating when considering classifying across rice cultivars that have similar phenotypes.

The failure of traditional methods to selectively direct attention to informative regions of an image is still another drawback. These techniques do not help to accentuate important grain patterns, forms and textures because they are non-selective for space or channel of origin as the small contemporary attention-based techniques are. This results into wrong predictions most especially when trying to distinguish between similar or related Group I rice cultivars.

In addition, what currently constitutes the best performers in the field, namely, Random Forests (RF) and Support Vector Machines (SVMs) also possess such limitations as inability to encode features of high levels of representation as well as the capacity to operate only with shallow information (He, Zhang, Ma, Li, & Ma, 2022; Sampaio, Castanho, Almeida, Oliveira, & Brites, 2020; Trisowati, Sartono, Kurnia, Dirgahayu, & Wijayanto, 2020). These models are slow and require more time and cannot accommodate large and diverse data sets as they often call for feature engineering. Despite classification task performance, they fail to learn hierarchical and multi-scale features of the input data.

On the other hand, using advanced deep learning (DL) techniques, such as CNN-based models, with attention mechanisms, these limitations are shifted as such models are capable of learning hierarchical nonlinear features simultaneously to focusing on channel and spatial domains. This leads to high level of accuracy and reliability in classification of rice variety irrespective of the complexity of feature data set.

In order to eliminate the shortcomings of the current state-of-the-art methods, this work introduces an Attention-Based Hybrid Model that considers both the channel and the spatial attention modes meanwhile using the CNN model to classify the rice varieties with high accuracy and robustness. However, traditional approaches like LDA, PCA, and manual feature extraction do not capture non-linear relations and minute visual patterns that arise from high dimension images of rice

grains. These approaches rely on ill-equipped, fixed linear transforms rather than being able to adaptively emphasize the most discriminative features and therefore offer deteriorated and non-uniform performance, especially for morphologically similar rice varieties.

The proposed Attention-Based Hybrid Model has been introduced to redress these challenges through new and improved attention mechanisms. For inference self-attention computational graph works adaptively on the channel dimension which helps to pay utmost attention to the concentration feature maps of texture, shape, and color. At the same time, the spatial attention module focuses on the features of rice grain images, or distinctive parts of the grain such as the pattern or texture while excluding other unnecessary features. These mechanisms coupled with the feature extraction hierarchical nature of the CNN resize the model to be sensitive to both high-level and low-level features making the classification more accurate. Due to the emphasis given to the most informative parts of the images, the proposed method accurately filters rice grain varieties which are quite similar in appearance, without the necessity of the manual definition of the features.

Key Objectives of the Study:

1. To Develop an Intelligent Classification Framework: Propose an Attention-Based Hybrid Model for channel and spatial attention to be incorporated in a CNN platform and improve the grain classification for rice.
2. To Address Limitations of Traditional Methods: Assess how the proposed model provides remedies to the limitations of selection techniques like LDA and ANOVA within feature selection for rice grain image analysis that lacks the ability to extract convoluted, non-linear data relations.
3. To Achieve Robust and Adaptive Performance: Apply the model on a wide range of data of different types of grains; rice grains especially exploring the ability of the model to scale to different texture, color, and shape related variations while having better accuracy.

Major Contributions of This Study:

1. A Novel Attention-Based Hybrid Model: Propose a model which combines both channel and spatial attention mechanism with the convolutional neural networks and makes the network focus on the discriminative features for rice variety classification.
2. Utilization of a Comprehensive Dataset: Consequently, this learn of models' performances by applying a large-scale dataset containing 20 kinds of rice grain both the original image and the augmented one.
3. Advancement in Feature Representation: Mitigate the drawbacks of conventional deterministic higher-order moment features of rice grains by integrating features capturing visual relations/interactions existing in huge and high-dimensional space of the images.
4. Scalable and Efficient Classification Framework: We recommend that the strategies create an efficient and repeatable assessment process that is more accurate and less labor-intensive than manual or conventional methods for achieving reliable quality assessment.

This work provides a new high-quality and reliable automated rice variety classification model which can satisfy the current global trend of demand for improving the recognition rate of automated agriculture and food quality measurement systems.

2. Literature review

This section delves into existing studies on visual datasets and rice variety classification in agricultural research. The literature review examines the key methodologies, datasets, and challenges addressed by previous works, focusing on rice varieties and classification frameworks.

2.1. Machine learning (ML) based models

Here, we will discuss several ML models used for rice variety classification. Saxena et al. (2022) introduced various ML approaches for rice variety classification employing a dataset comprising five globally grown rice varieties with 75,000 samples with 15,000 per class, and 107 features were investigated, where the top 20 features were selected employing the RF Classifier. Several ML classifiers such as Logistic Regression (LR), Decision Tree (DT), SVM, RF, Perceptron, KNN, and Gaussian Naïve Bayes (GNB) were trained on 70%–30% and 80%–20% data splits. Outcomes revealed notable advancements with feature selection, where RF performed the highest accuracy of 99.85%.

Cinar and Koklu (2022) classifying five rice varieties using morphological, shape, and color features by employing 75,000 rice grain images as 15,000 per variety were pre-processed utilizing MATLAB for feature extraction. These images provided 106 features, including 90 color features from five color spaces, four form features, and 12 morphological traits. Various ML algorithms, such as KNN, DT, LR, multilayer perceptron, RF, and SVMs, were applied to evaluate different feature sets (12, 16, 90, and 106 features). The outcomes showed that RF obtained 98.04% accuracy with morphological and form features and 97.99% accuracy with morphological features.

Qadri et al. (2021) used machine vision (MV) techniques to classify six Asian rice varieties cultivated in Pakistan, China, India, and Bangladesh. The dataset has 10,800 rice grain samples, 1800 per variety captured in the field using a cell phone camera. A feature vector of 154,800 was obtained by converting the images to 8-bit grayscale, extracting six non-overlapping regions of interest, and then extracting 43 features (binary, histogram, and texture). Classification accuracies of 97.4%, 97.0%, 96.3%, 95.74%, and 95.2% were attained by applying five MV classifiers: LMT Tree (LMT-T), Meta Classifier via Regression (MCR), Meta Bagging (MB), Tree J48 (T-J48), and Meta Attribute Select Classifier (MAS-C).

Ruslan, Khairunniza-Bejo, Jahari, and Ibrahim (2022) presented image processing and ML techniques to classify weedy rice seed variants and cultivated rice seeds. A machine vision unit took a weedy rice seed variety and five cultivated rice kinds in RGB and monochrome. These photos were analyzed to extract 67 morphological, color, and texture variables fed into ML classifiers. Performance measures assessed seven classifiers: accuracy, sensitivity, specificity, and average correct classification. The LR model, with 85.3% sensitivity, 99.5% specificity, 97.9% accuracy, and 92.4% average correct classification, performed best when RGB images were used, according to the results.

To classify different rice types, Naik, Subbarao, Sethy, Behera, and Panigrahi (2024) presented 75,000 rice grain images, with 15,000 images for each variety. Combining 12 morphological, four shapes, and 90 color features extracted from five color spaces resulted in 106 total features. An analysis of variance (ANOVA) was used to select high-rank features, which were then input into an SVM for classification. The results demonstrated that using 40 high-rank features led to impressive performance, achieving 99.89% validation accuracy and 99.81% test accuracy, with cost values of 16 and 28, respectively.

Kang et al. (2024) classified five types of rice, including Thai jasmine rice, by examining their characteristics in the 475–1000 nm wavelength range using fluorescence hyperspectral technology and ML. Before feature reduction using PCA and t-SNE, the data were preprocessed with a first-order derivative (FD) to minimize baseline drift. Classification models such as BP neural network, PLS-DA, and RF were used. The gray wolf algorithm (GWO) was employed to optimize RF with a training accuracy of 99.8% and a testing accuracy of 95.3%.

2.2. Deep learning (DL) based models

A Deep Convolutional Neural Network (DCNN) technique called RiceNet was developed by Din et al. (2024) to categorize five different types of rice. With a learning rate of 0.00003, the Adam optimizer was

used to fine-tune pre-trained models such as InceptionV3 and InceptionResNetV2. Traditional machine learning models, such as HOG-SVM, SIFT-SVM, HOG-LR, SIFT-LR, HOG-KNN, and SIFT-KNN, were compared to RiceNet's performance. InceptionV3 and ResNetInceptionV2 achieved 84% and 81.33% accuracy, respectively, whereas RiceNet outperformed the other models with an accuracy of 94%.

Sharma, Sethi, and Bawa (2024) presented advanced DL and transfer learning (TL) classifiers for rice variety detection, utilizing a large dataset of over 600,000 rice images divided into 22 classes to improve classification accuracy. Among the DL models tested, ResNet50 demonstrated the best performance with 96% training accuracy and 80.5% testing accuracy. Other models compared in the study include CNN, Deep CNN, AlexNet2, Xception, Inception V3, and DenseNet121.

A hybrid DL model for rice variety classification was presented by Mamun, Karim, Sarkar, and Alam (2024), using a heterogeneous dataset of 20,000 high-resolution RGB representing five major rice varieties. To solve the vanishing gradient issue, the model design included a convolutional layer, zero-padding, batch normalization, max pooling, and residual blocks with an SVM model. Furthermore, two well-known models optimized for the task, MobileNetV2 and VGG16, were compared for performance. The hybrid model outperformed VGG16 95% and MobileNetV2 93% in precision, recall, and F1-score measures, achieving remarkable results with 99% accuracy.

RiceSeedNet, a deep neural network architecture paired with convolutional image processing methods, was presented by Rajalakshmi, Faizal, Sivasankaran, and Geetha (2024) for the classification of regional rice seed varieties from southern Tamilnadu, India. There are 13,000 RGB photos in the RiceSeed Image corpus, with 1000 images for each of the 13 local rice seed variants. A remarkable 97% classification accuracy was attained by the RiceSeedNet model when it came to local rice seed variants. The model also showed 99% accuracy when tested on a publicly available dataset of rice grains.

Taheri, Ebrahimnezhad, and Sedaaghi (2024) presented a unique approach for the high-precision hyperspectral image (HSI) classification of 90 rice seed classes using ensemble DL approaches. The chosen hyperspectral and RGB information from rice seed photos is used to train a deep neural network, producing distinct models for every band. Lastly, an ensemble technique improves classification accuracy by combining these models. With just 15 chosen hyperspectral bands, the approach achieves remarkable precision ranging from 92.73% to 96.17%, despite the difficulties of numerous classes and few data samples per class.

Jeyaraj, Asokan, and Nadar (2022) designed a precise DL-based rice grading system that uses a NI-myRIO with a high-resolution camera and user interface to process rice in real-time based on its appearance and attributes. The network was first trained to extract discriminative features from a publicly available rice dataset. Two publicly available standard datasets and a real-time scanning prototype built using the AlexNet architecture were used to evaluate the suggested DL model, and the average accuracy was 98.2%, with a sensitivity of 97.6% and a specificity of 96.4%.

A suitable method for developing an automated system for rice recognition was suggested by Hoang, Van Hoai, Surinwarangkoon, Duong, and Meethongjan (2020). A range of CNN techniques and manually generated descriptors are assessed and contrasted. The experiment was carried out using the VNRICE dataset, and the suggested approach produced noteworthy outcomes using the DenseNet21 framework of 99.04%.

Olusegun (2024) presented a novel hybrid approach integrating CNNs and RNNs to capture spatial and temporal features from rice images. The CNN retrieves rich feature representations, whereas the RNN records contextual information and sequential dependencies. With an accuracy of 95%, the hybrid models perform noticeably better than traditional and single DL models, according to tests conducted on a large dataset of several rice types in various environmental circumstances.

After reviewing the entire literature, we discover some notable limitations:

Table 1
Overview of the existing studies discussed earlier.

Ref	Dataset	Dataset size	Rice variety	Classifiers	Performance
Saxena et al. (2022)	Rice image dataset	75,000	05	RF, LR, DT, SVM, Perceptron, KNN, NB	99.85%
Cinar and Koklu (2022)	Rice image dataset	75,000	05	KNN, DT, LR, multilayer perceptron, RF, SVMs	99.91%
Qadri et al. (2021)	Rice image dataset	10,800	06	LMT-T, MCR, MB, T-J48, MAS-C	97.04%
Ruslan et al. (2022)	Manually collected	7350	05	DT, NB, LR, KNN, Discriminant Analysis, Ensemble	97.9%
Naik et al. (2024)	Manually collected	15,000	05	SVM	99.89%
Kang et al. (2024)	The fluorescence hyperspectral data	550	05	FD-t-SNE-GWO-RF	99.8%
Din et al. (2024)	Manually collected	4748	05	DCNN	94%
Sharma et al. (2024)	–	600,000	22	InceptionV3, ResNetInceptionV2	80.5%
Mamun et al. (2024)	Manually collected	20,000	05	CNN+SVM	99%
Rajalakshmi et al. (2024)	RiceSeed	13,000	13	RiceSeedNet	99%
Jeyaraj et al. (2022)	Kaggle	750	05	AlexNet	98.2%
Hoang et al. (2020)	VNRICE	10,508	06	DenseNet21	99.04%
Olusegun (2024)	Manually collected	–	–	CNN+RNN	95%

- **Inconsistent Visual Inspection:** The traditional visual inspection method depends heavily on human experience, which can lead to subjective and conflicting results. Inspectors may interpret visual cues differently visible cues, leading to errors in classification.
- **Limited Labeled Data:** Traditional classifiers require much-labeled data for training. However, obtaining sufficient labeled samples for various rice varieties can be challenging, limiting the effectiveness of these models.
- **Complex Rice Samples:** Rice varieties can have slight differences that are difficult to detect, such as slight variations in size, shape, or color.
- **Feature Extraction Challenges:** Extracting the right features from rice grains can be challenging.
- **Overfitting in Machine Learning Models:** Models may overfit the training dataset if there is a lack of labeled data, resulting in poor generalization of new, untested data.
- **Environmental Factors:** Rice varieties may exhibit different characteristics under varying ecological conditions, making classification more complex and inconsistent. Additionally, Lighting conditions may affect the accuracy of rice variety classification. Insufficient brightness may cause overexposure by allowing shadows to determine various types. Inaccurate categorization findings may arise from distracting photo backdrops that interfere with the attributes taken from rice grains.
- **Computational Complexity:** Certain sophisticated classification models, especially DL models, could need much time and computing power to train, which is not always possible in academic or commercial contexts (see Table 1).

3. Methodology

An advantage in the quick advancement of digital and artificial intelligence technology has propelled the innovation of automatic systems for agro-analysis improvements. Especially for farming industry which is a main resource in Bangladesh, mystery of different types of rice is crucial necessity for right selection of crop and its management, increased production, quality of food and research in agriculture. Actually analyzing the flow of instructions is a part of the traditional approach that involves manual inspection — a slow and error-priddled process. Employing the spatial variables such as size and texture of grains, and shape deep learning techniques especially CNNs has been enhanced to be effective tools in automating tasks of categorization.

There is still the issue of trying to distinguish between rice classes with high variation within each class while having little variation between each class, even if this much improvement has been evidenced by CNNs. To enhance feature extraction and classification of dopaminergic neurons in human midbrains, this work developed a new CNN model incorporated with the CBAM.

3.1. Dataset of Bangladeshi rice varieties

This research provides a rich photographic archive of rice types from Bangladesh, and has been collected from Mendeley Data, Aruzz22.5K: An Image Dataset of Rice Varieties, published on 12 March 2024 (Islam, 2024), consisting of twenty distinct grain types documented in twenty-seven thousand photos. BR28, BR29, Paijam, Bashful, Lal Aush, Jirashail, Gutisharna, Birui, Najirshail, Pahari Birui, Polao (Katari), Polao (Chinigura), Amon, Shorna, Subol Lota, Bashmoti, Ganjiya, Shampakatari, Sugandhi, Katarivog are such types of grains that comprise these classes. The overview of the dataset is presented in Table 2.

The selected dataset provides the systematic analysis of these rice cultivars including their significant morphological characteristics for right classification. To mimic actual-world variations in grain morphology, the photographs were attained using iPhone 11 camera with a 5x zoom functionality starting from October 18 to November 29 of 2023. Different classes of rice grains are shown in Fig. 1. Each row represents a distinct grain variety to highlight the visual differences.

3.1.1. Dataset composition

The dataset is divided into two main parts:

1. **Original Image Dataset:** The high-resolution images in the original dataset comprise of 4730 numbers. All the images presented on this page are of the same dimensions, 853 × 853 pixels, and all were shot in natural light; the author captured several types of rice in these photos. The dataset used for current study has been carefully chosen to emphasize the textural properties such as texture pattern, shape and grain size. This dataset is conditionally homophilous because the original photos were taken with an iPhone 11 camera.
2. **Augmented Image Dataset:** An additional 23,650 images were synthesized using data augmentation approaches to enlarge and enhance the dataset variety. These are standard 512 × 512 pixel pictures that are enlarged to provide a range of morphological

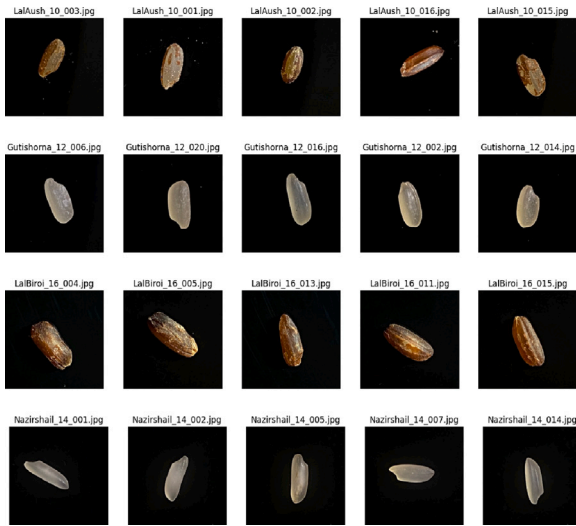


Fig. 1. Visual representation of different rice grain varieties.

Table 2
Bangladeshi rice varieties dataset details.

Aspect	Details	Notes
Dataset name	Bangladeshi rice varieties dataset	Includes 20 rice varieties.
Classes	20 rice varieties	Includes Subol Lota, Bashmoti, Ganjiya, Chinigura Polao, etc.
Original image dataset	4730 images	853 × 853 pixels, iPhone 11, original quality.
Augmented image dataset	23,650 images	512 × 512 pixels, through data augmentation.
Image capture period	October 18–November 29, 2023	Captured in rural and marketplace settings.
Image format	JPG	Uniform file format across the dataset.
Data organization	20 subdirectories	Each subdirectory corresponds to a rice variety.
Train set details	1125 images per class	Stored in the '1_TEST' folder.
Validation set details	225 images per class	Stored in the '2_VALID' folder.
Total image count (Feed to Model)	27,000 images	Includes both original and augmented images.

changes such as enlargement, flipping, and rotation. With such variations, the deep learning models can train with a more generalized set of features in an augmented dataset.

3.2. Preprocessing and data augmentation

This paper has also explored the methods of data preprocessing and augmentation to ensure high quality of inputs for the classification and improving the performance based on the Attention-Based Hybrid Model to classify rice variety. Since data pre-processing was important in unification and the right feature extraction, images were sized to 512 × 512 pixels. The images resized into 28 × 28 pixels for computational reasons at some stages and flattened into a one-dimensional array of shape (28 × 28 × 3). Flattening especially maintains intricate figure qualities while ensuring compatibility with dense and numerous layers because the image transition into a linear format.

The following method was used to normalize pixel values to the range [0, 1]:

$$I_{\text{norm}} = \frac{I_{\text{raw}}}{255} \quad (1)$$

Normalization improves overall learning efficiency by stabilizing the training process, preventing excessive gradients, and speeding up convergence.

Data augmentation methods like rotation, flipping, scaling, and brightness alteration were used to improve generalization and lessen overfitting:

$$I' = T(I), \quad T \in \{\text{rotation, scaling, flipping, brightness adjustment}\} \quad (2)$$

The model may learn invariant and discriminative features thanks to this variability, which also makes it resistant to changes in lighting, scale, and orientation.

Each rice variety was given a unique integer identifier using label mapping (for example, 1_Subol_Lota mapped to 1), and then one-hot encoding into binary vectors of size C (where $C = 20$) was performed:

$$y = [y_1, y_2, \dots, y_C], \quad y_i = \begin{cases} 1 & \text{if } i = \text{class index,} \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

This kind of model is highly advantageous because of the certainty of achieving as clear and accurate a categorization as possible.

In essence, standardization makes training less erratic, while resizing and flattening ensure fast computations, and homogeneous inputs. While labeling mapping enables separation of classes while augmenting enhances the models ability to address a range of changes. In combination, these actions make it possible for the Convolutional Block Attention Module (CBAM) to convene and focus on channel prospects and influential spatial areas in an efficient manner and enhancing model's ability to categorize Bangladeshi rice types precisely and reliably.

3.3. Proposed attention-based hybrid model

Our proposed approach includes an Attention-Based Convolutional Neural Network that seeks to improve feature extraction across the architecture by focusing attention on important parts of the data. The self-attention mechanism pays more attention not only to the most informative spatial features but also enhances the representation of the complex structure. Finally, the features extracted by the Attention-Based CNN are input to some of the basic classification models including RF, SVM, LR, KNN, XGBoost, and MLP. The overall proposed model is shown diagrammatically in Fig. 2.

3.3.1. CNN block

The CNN block is followed to derive hierarchical spatial features from the incoming data. It performs convolution, activation, pooling, and normalization sequentially to guarantee that the network surface can capture fine spatial patterns successfully.

- **Convolutional Layers:** This layers identify shapes such as edges or textures, by feeding the input X through learnable filters, also called W . The mathematical operation is defined as:

$$Z = \text{ReLU}(\text{Conv}(X, W) + b) \quad (4)$$

where, X is the input tensor, W represents the kernel/filter, b is the bias term, $\text{Conv}(X, W)$ is the convolution operation and $\text{ReLU}(z) = \max(0, z)$ introduces non-linearity. This step enables the network to incorporate elaborate spatial characteristics into its modeling process.

- **Max Pooling:** This layer shrinks the spatial dimensions down to the most important existing within the sliding window. And the operation ensures computational efficiency and reduces sensitivity to spatial translations:

$$P = \text{MaxPool}(Z) \quad (5)$$

where, Z is the output from the previous layer (after convolution and ReLU activation) and P is the output after applying the max pooling operation.

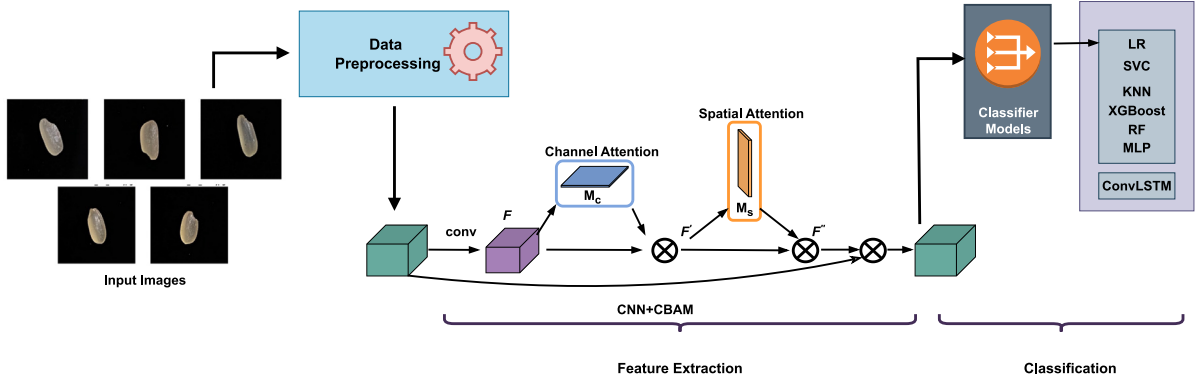


Fig. 2. Proposed models workflow.

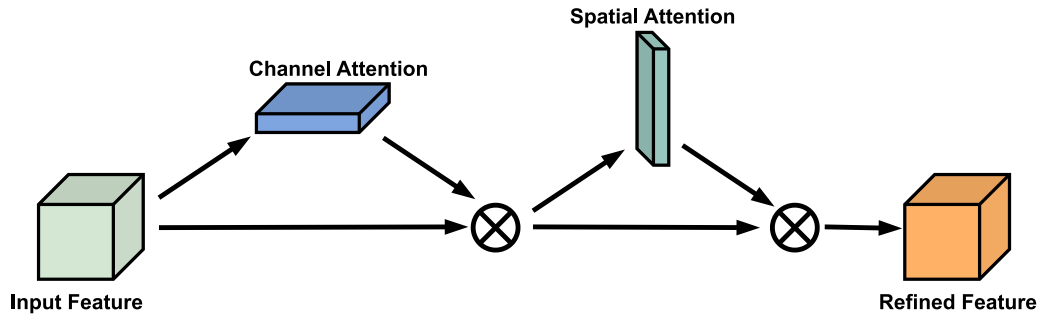


Fig. 3. Structure of CBAM.

- **Batch Normalization:** Reduces the variations in activations for maintaining stable distributions during training of network. The formula for normalization is:

$$\hat{Z} = \frac{Z - \mu}{\sqrt{\sigma^2 + \epsilon}} \quad (6)$$

where, μ is the mean, σ^2 is the variance and ϵ prevents division by zero.

The model becomes more resilient and converges more quickly during training by combining pooling and normalization.

3.4. Attention block

The CBAM attention mechanism is actually made up of two modules, the Channel Attention Module (CAM) as well as the Spatial Attention Module (SAM). These modules operate to control spatial and channel attention. The advantage of CBAM is in its capacity for the elaborate examination of the importance of components in spatial and channel fields, differing from other phenomenal attention processes (Chen & Dang, 2023). More precisely, CBAM improves the CNN by preserving only the most important features for both channel and spatial dimensions. The general block diagram for CBAM is visualized in Fig. 3.

3.4.1. Channel attention module (CAM)

Channel Attention, states that the “what” features are important and redistributes channel weights for pertinent patterns. Actually, This module calculates the relevance of each channel by using the Global Average Pooling and Global Max Pooling across the spatial dimension. These outputs are then feed into a shared MLP containing two dense layer, whose outputs are summed element-wise and passed through a sigmoid activation function to generate the channel attention map M_c . The channel and spatial attention module is shown visually in Fig. 4.

- **Global Average and Max Pooling:** These operations put together data from the spatial aspect for each of the channels. Let F denote the input feature map. Compute channel-wise descriptors:

$$F_{\text{avg}} = \text{AvgPool}(F) \quad (7)$$

$$F_{\text{max}} = \text{MaxPool}(F) \quad (8)$$

- **Shared MLP:** This layer uses a shared Multi-Layer Perceptron (MLP) to process pooled features:

$$M_{\text{avg}} = \text{MLP}(F_{\text{avg}}) \quad (9)$$

$$M_{\text{max}} = \text{MLP}(F_{\text{max}}) \quad (10)$$

- **Element-wise Summation and Sigmoid Activation:** Using a sigmoid activation, the outputs of average and max pooling are combined to generate the channel attention map.

$$M_c = \sigma(M_{\text{avg}} + M_{\text{max}}) \quad (11)$$

- **Element-wise Multiplication:** Refine the input feature map:

$$F' = M_c \odot F \quad (12)$$

where, $\odot$ represents element-wise multiplication.

3.4.2. Spatial attention module (SAM)

This module assesses the relevance of spatial areas. It utilizes average as well as max pooling over the channel dimension to produce two feature maps. These are concatenated and feed through a $7 * 7$ convolutional layer to generate a spatial attention map. This map enhances crucial spatial areas in the input feature maps. By emphasizing significant spatial regions, the SAM concentrates on determining “where” on the feature map F' . It uses the CAM-refined feature map as input.

- **Global Avg and Max Pooling:** Utilize pooling along the channel axis to calculate descriptors:

$$F_{\text{avg}}^{\text{spatial}} = \text{AvgPool}(F') \quad (13)$$

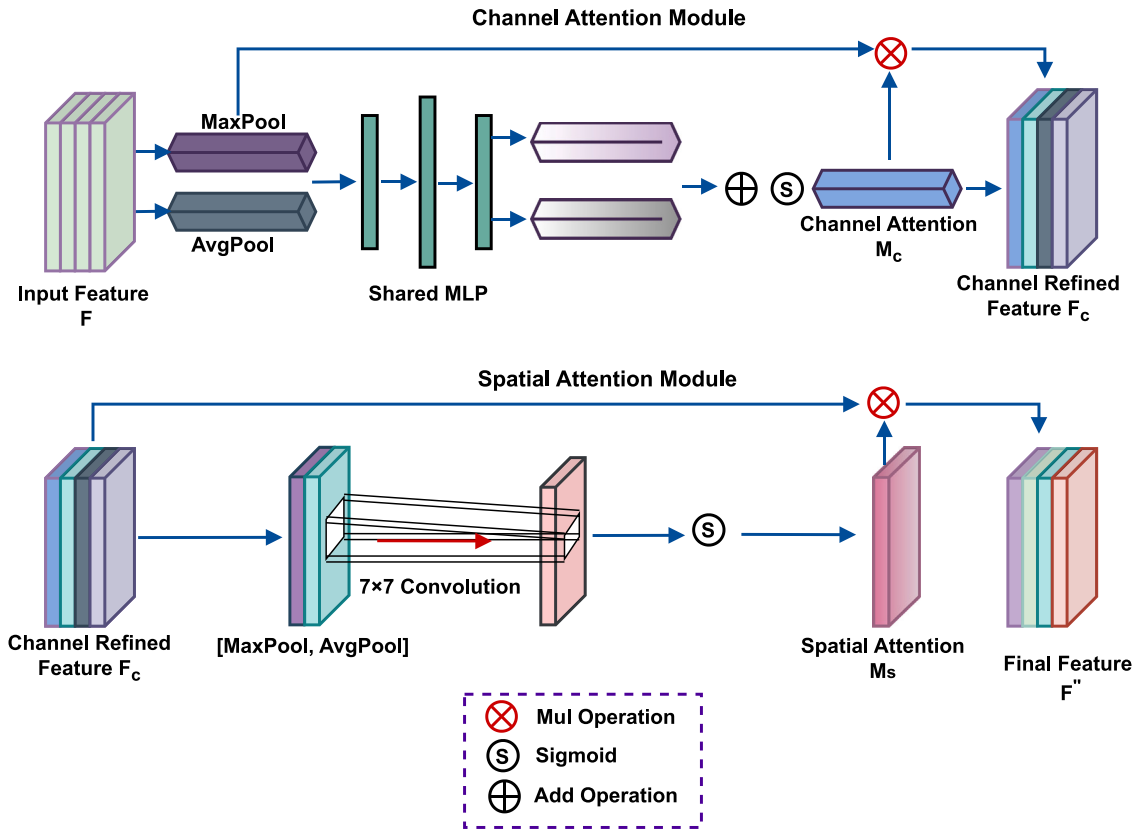


Fig. 4. Structure of CAM and SAM.

$$F_{\max}^{\text{spatial}} = \text{MaxPool}(F') \quad (14)$$

- **Channel Concatenation:** To create spatial attention weights, this layer concatenate the pooled descriptors:

$$F_{\text{concat}} = \text{Concat}(F_{\text{avg}}^{\text{spatial}}, F_{\max}^{\text{spatial}}) \quad (15)$$

- **Convolution and Sigmoid Activation:** Apply a 7×7 convolution and a sigmoid activation to generate the spatial attention map:

$$M_s = \sigma(\text{Conv}(F_{\text{concat}})) \quad (16)$$

- **Element-wise Multiplication:** Refine the input feature map:

$$F'' = M_s \odot F' \quad (17)$$

Integrating both CAM and SAM mechanisms, the introduced CBAM module enhances the representation capability of features that are indispensable in the feature map. Specifically, the CAM improves channel-wise relevance, while the SAM focuses on spatially important areas, making the CBAM an optimal lightweight attention scheme for diverse applications of deep learning. The Attention-based CNN model block is illustrated in Fig. 5.

3.4.3. Classifier models

The features extracted from the input data and after training the Attention-Based CNN model is used for the classification purpose. To focus on the important part of the feature map and attenuate the less important part, the model uses two attention mechanisms: CAM and SAM. These features are then passed through a baseline classifier that forms the last stage of feature-based classification. The classifier makes use of the extended feature set so that it is able to perform the task of separating as well as identifying the different rice varieties and sub varieties and so on in an efficient and effective manner. The integration of these attention mechanisms with the baseline classifier helps

in a remarkable enhancement of the model's capacity to handle the Rice variety classification task. The following models were employed for classification: The tested algorithms include Random Forest (RF), Support Vector Classifier (SVC), Logistic Regression (LR), K-Nearest Neighbors (KNN), XGBoost, Multilayer Perceptron (MLP), ConvLSTM (based on the initial images).

1. **Random Forest (RF):** The RF is a hierarchical clustering of the base classifiers based on a tree architecture of decision trees. To determine one property with the greatest relevance, the RF method uses a simple precondition probability. On that basis, Breiman further developed the RF technique by mapping a random sample of feature subspace to build several decision trees by using sample data subsets (Jackins, Vimal, Kaliappan, & Lee, 2021) and can be expressed as:

$$\{DT(x, \theta_k)\}_{k=1}^T \quad (18)$$

where x is the input vector, and θ_k denotes a random vector, which is sampled independently but with the same distribution as the previous $\theta_1, \dots, \theta_{k-1}$. T bootstrap samples are initially derived from the training data (Magidi, Nhamo, Mpandeli, & Mabhaudhi, 2021).

In our study, we have utilized $n_{\text{estimators}} = 100$, meaning that there were 100 decision trees used in the development of the ensemble. This selection aims at finding an optimal trade-off between the computational time and classification accuracy. In general, more estimators lead to higher stability and lower variance as well as longer training time. This tradeoff was beneficial for our classification task because it gave accurate as well as stable results using 100 estimators.

2. **Support Vector Classifier (SVC):** SVC is a classical linear classifier with highly capability to perform generalization. The principle of the structural risk minimization forms the basis of the

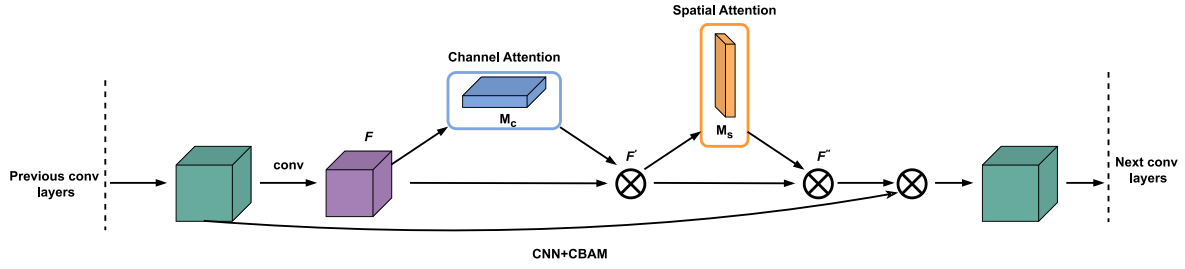


Fig. 5. Structure of attention-based CNN model block.

classifier. This one partitions the samples with the help of a maximum edge hyperplane and then transfer the data to a high dimensional eigenspace (Li et al., 2021). The objective of the SVC can be expressed mathematically as:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{subject to} \quad y_i(w \cdot x_i + b) \geq 1, \quad \forall i \quad (19)$$

where, x_i represents the input feature vector, $y_i \in \{-1, 1\}$ denotes the class label, w is the weight vector defining the hyperplane, b is the bias term, and $\|w\|^2$ is the regularization term to ensure maximum margin. SVC transforms the input data into a higher level n -dimensional space and uses kernel functions to make data to be separable non-linearly. This makes it possible for the hyperplane to better distinguish between classes than its corresponding feature. The kernel trick makes sure that computational requirement is optimized by introducing the optimization issue without having to compute the map into a high-dimensional feature space.

3. **Logistic Regression (LR):** The way for applying the linear regression to classification issues is offered by this technique. In practice, the above classification extends yields a result falling within the range $[0, 1]$, can be regarded as probabilistic density function $h(x)$, which indicates that x belongs to class 1. Namely, the function considered to be the logistic function and defined as follows is the sigmoid function for logistic regression:

$$f(z) = \frac{1}{1 + e^{-z}} \quad (20)$$

where z is of the form, $z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$, where x_1 to x_n represent the values of the n attributes and β_0 to β_n represent the weights (Guerrero, Parada, & Espitia, 2021).

4. **K-Nearest Neighbors (KNN):** Another kind of intelligent algorithms which do not contain the learning stage are known as KNN algorithms. Its base is the calculation of the distances between the sampling points and the accumulation of the closest neighbors of the assigned points. Since then it has continued as above being based on the majority vote of the KNN (Kherif, Benmahamed, Tegar, Boubakeur, & Ghoneim, 2021). kNN algorithm can employ a range of distance measures such as cosine, Manhattan, Euclidean and others in order to come up with the distance estimate of an object. Most commonly used distance is the Euclidean distance; it is frequently implemented into the model defined as:

$$d(z_i, z) = \left(\sum_{l=1}^q (z_{li} - z_l)^2 \right)^{1/2} \quad (21)$$

where z_{li} and z_l are the values of the l th feature of objects z_i and z , respectively, and q is the number of features (Demidova, 2021).

5. **XGBoost:** Chen and Guestrin introduced XGBoost, a gradient boosted decision tree model. The XGBoost technique sequentially trains decision trees using training data. To increase the value of the goal function, the algorithm appends a new decision tree to the earlier decision trees at each iteration. The loss term (l) and the regularization term Ω make up the objective function,

which is meant to be minimized. The objective function of the t -th iteration (L_t) is defined by Eq. (22), where n is the number of instances in the training set, f_k is the function of the tree, Ω is the regularization term, y_i is the actual class label of instance i , and $\hat{y}_i$ is the predicted class label of instance i . The objective function, which is aimed to be minimized, consists of the loss term (l) and the regularization term (Ω). The objective function for the t th iteration (L_t) is defined as:

$$L_t = \sum_{i=1}^n l(y_i, \hat{y}_i^{t-1} + f_t(x_i)) + \Omega(f_t) \quad (22)$$

where: y_i is the actual class label of instance i , $\hat{y}_i^{t-1}$ is the predicted class label of instance i from the previous iteration, $f_t(x_i)$ is the function of the current decision tree, n is the number of instances in the training set, $\Omega(f_t)$ is the regularization term (Aydin & Ozturk, 2021).

The regularization term $\Omega(f_t)$ penalizes model complexity to avoid overfitting. It is defined as:

$$\Omega(f_t) = \gamma T_t + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (23)$$

6. **Multilayer Perceptron (MLP):** Among the classifiers for classification process, the Multi-layer Perceptron Classifier (MLP Classifier) uses an underlying Neural Network. The MLP algorithm is carried out by the MLP Classifier but the algorithm uses backpropagation to train the neural network (Goh & Yann, 2021). MLPs are neural networks that use dynamic classifiers. The decision-making in the MLP classifier is done by the use of neural network and arrives at a hyperplane for separating different cherry types. The MLP mainly consists of an input layer, a hidden layer and an output layer and each layer is connected with the neighboring layer in the entirely (Song, Yang, & Wang, 2023). To measure the effectiveness of the network during training a loss function is used. The common one used is mean squared error as shown in following:

$$L(y, \hat{y}) = \frac{1}{2} (y - \hat{y})^2 \quad (24)$$

where, y is the actual value and $\hat{y}$ is the predicted value.

7. **ConvLSTM:** To take the advantages of both CNNs and LSTM for classification we used a ConvLSTM model in the present study. CNN component employs non-linearity, max-pooling and consecutive convolution to extract adequate spatial features from the input vision. Meanwhile, these spatial information are being temporally processed in LSTM component where dependencies of the data sequences such as video frames are held. For multi-class classification, the hybrid approach on linking CNN and LSTM is performed by concatenating the final layer's outputs and feed the data into the fully connected dense layers with softmax activation. It provides a strong model for addressing spatiotemporal data management by integrating the sequence prediction ability of LSTMs with the CNN spatial feature extraction ability.

Table 3
Class division with a split of 75:25.

Class division	Train	Test
1_Subol_Lota	1013	337
2_Bashmoti	1013	337
3_Ganjiya	1012	338
4_Shampakatari	1013	337
5_Katarivog	1012	338
6_BR28	1013	337
7_BR29	1012	338
8_Paijam	1012	338
9_Bashful	1013	337
10_Lal_Aush	1012	338
11_Jirashail	1012	338
12_Gutisharna	1013	337
13_Red_Cargo	1012	338
14_Najirshail	1012	338
15_Katari_Polao	1012	338
16_Lal_Biroi	1013	337
17_Chinigura_Polao	1013	337
18_Amon	1013	337
19_Shorna5	1013	337
20_Lal_Binni	1012	338

The LSTM layer mathematically represents sequential dependencies as follows:

$$h_t = \sigma(W_x x_t + W_h h_{t-1} + b) \quad (25)$$

where, h_t is the hidden state at time step t , x_t is the input vector at time step t , W_x is the weight matrix for the input, W_h is the weight matrix for the previous hidden state, b is the bias vector, and σ is the activation function.

4. Experimental setup

4.1. Implementation and splitting

In our implementation phase of the designed experimental setup, Google Colab Pro+ has been used having a specific runtime type set that uses the A100 GPU to achieve better computational power. The model training process was done in roughly 1407.936 s of time which again emphasizes the utility of GPU when working with large numerations and calculations. The development of the implementation was conducted on Python which was the main language used in the process to enhance ability to develop hence making the process flexible. Subsequently, for label mapping, the 'os' using 'os.path.join()' function has been used to combine image path with corresponding label. Moreover, 'Image' function from Python's PIL was used for image processing. Especially, the 'image_to_pixels()' function was created to read images with the help of the 'Image.open()' program and convert images into an array of pixels for the subsequent analysis. These tools and configurations optimized the working process and made data preprocessing, model training and evaluating process fast and efficient for the rice classification task.

The data set was divided into train and test set in training process using the function `train_test_split` of Scikit-learn. For the distribution of classes, the data set was split in to training as well as test data into 85:25 proportion. The detailed class distribution is given using Table 3. When splitting the data, it was necessary to use the parameter of `random_state` with value as 42 to compare results with different runs. Also, the 'stratify=Label_encoded' function was used with a view of emulating the class distribution ratio in the training and testing sets as was with the original data. This stratification is important in ensuring that all classes are represented equally within a split both when a dataset has issues with class imbalance and this would make the evaluation of the model much more reliable.

Table 4
Layers and parameters in the CBAM module.

Component	Details
Input Shape	(28 × 28 × 3)
Channel Attention Layers	2 Dense, 1 GlobalAveragePooling2D, 1 GlobalMaxPooling2D
Spatial Attention Layers	1 Conv2D, 1 Lambda for Average Pooling, 1 Lambda Layer for Max Pooling, 1 Concatenate Layer
Residual Connection Layers	2 Add and Multiply layers
Reduction Ratio	16
Dense Activations	ReLU (Layer 1), Sigmoid (Layer 2)
Dense Kernel_INITIALIZER	he_normal
Conv2D Kernel Size	7 × 7

4.2. Hyperparameters

4.2.1. Hyperparameter configuration for CBAM layer

CBAM is another important addition for fine tuning the rice classification model by integrating channel and spatial attention mechanisms. The CBAM implementation enables the hyperparameters to be tuned, including the reduction ratio in the channel attention part. This reduction ratio has been fixed at 16 as this is the almost ideal midpoint between having a higher dimensionality reduction while not losing much information on the features. In addition, the Fully connected layers for channel attention are normalized using He Normal to ensure gradients do not distribute poorly, while biases are set to zero for uniformity. For spatial attention, to extract the more complex features of spatial relatedness, a convolution of 7 × 7 and sigmoid activation function is used. These design choices make sure that the CBAM successfully emphasize informative regions within the feature maps while the training is in progress (see Table 4).

Every hyperparameters used in our work is thoroughly described here.

1. **Activation Function:** In artificial neural networks, activation functions are used exclusively to map an input signal into an output signal for the subsequent layer at the next stratum level. Any artificial neural network adopts an activation function once the summation of products of inputs and their corresponding weights is computed (Sharma, Sharma, & Athaiya, 2017). The ReLU activation function, which is one of the most popular activation functions in neural networks and is well-known for solving the vanishing gradient issue and cutting down on the sluggish training time of saturated S-type activation functions (Qiumei, Dan, & Fenghua, 2019), is employed in CBAM class. The following is its mathematical expression:

$$f(x) = \begin{cases} x_i, & \text{if } x_i \geq 0 \\ 0, & \text{if } x_i < 0 \end{cases} \quad (26)$$

We employed the sigmoid activation function in Layer 2, which calculates the response for a given signal at point x as follows:

$$f(x) = \frac{1}{1 + \exp(-x)} \quad (27)$$

This function has an S-shaped curve and spans from 0 to 1. Despite being simple to use, the sigmoid function has a number of drawbacks. Notably, the gradients have a tendency to converge to zero during training, and it can result in sluggish convergence in very deep networks. Gradients are essential to learning and weight updates, and the network performs poorly as they go close to zero (Khalid et al., 2020).

2. **Reduction Ratio:** Our study's channel attention mechanism uses a multi-layer perceptron (MLP) to calculate channel-wise attention scores after compressing the input feature maps along the channel dimension using techniques like global average pooling or global max pooling. The degree of this compression is determined by the `reduction_ratio`. As a result, the network's capacity

Table 5
Hyperparameter setup for the attention-based CNN model.

Hyperparameter	Value
Input shape	(28, 28, 3)
Filter sizes	32, 64, 128, 256
Dropout rate	0.2
Activation (Hidden layers)	ReLU
Output activation	Softmax
Loss function	categorical_crossentropy
Output kernel initializer	glorot_uniform
Number of classes	20
CNN Layers	7 Conv2D, 4 MaxPooling2D, 7 BatchNormalization and 4 Dense layers
Dense layer units	256, 128, 64, 32
Optimizer	Adamax
Learning rate	0.001
Verbose	1
Batch size	32
Epoch	100

to highlight important characteristics is especially impacted by the reduction ratio.

- Kernel Initializer:** The `kernel_initializer` variable defines how the weights are initialized (Guo, 2023). Also `he_normal` is a method of kernel initialization which is described in He, Zhang, Ren, and Sun (2015), which is based on Glorot and Bengio (2010), and performs faster convergence of deeper CNNs. In forward propagation case, the central idea of `he_normal` is to examine the variance of the responses in each layer and the strategy of the initialization should not reduce the input signals magnitudes exponentially (Naseer & Saleem, 2018).

4.2.2. Hyperparameter setup for the attention-based CNN model

The Attention-based CNN model incorporates the CBAM into the deep convolutional neural network designed optimally for accurate rice variety classification. These include; the number of convolution layers, dropout rates and the number of neurons in the dense layer, type of optimizer used. The convolutional blocks are progressively constructed wherein the filter of the network increases from 32 to 64, 128 and 256 respectively to extract hierarchical features. Each block apply Batch Normalization to accelerate the convergence speed while use MaxPooling to reduce the feature dimension. To combat overfitting, the dropout layers are applied with a dropout of 20% for the fully connected layers. The first type of dense layers include units in the layer decreasing from 256 to 32 then ReLU activation, followed by Batch Normalization to enhance the learning process. The final layer of classification uses the Softmax function to give a probability of each rice class that is being classified (see Table 5).

- Dropout Rate:** The probability that the neuron will be kept is represented by the dropout rate. On the one hand, using Dropout to avoid overfitting lowers the number of network parameters used in the computation each time the gradient is changed, which lowers the model's capacity; on the other hand, it causes weight dispersion, which is a regularization technique (Zhao, 2024). In the current study, a dropout rate of 0.2 was used in order to strike this balance.
- Loss Function:** Variable values are mapped onto a real number via the loss function, which intuitively indicates some loss related to the variables. Furthermore, the associated loss in neural network-based multi-classifier problems is the discrepancy between the activation function's output value and the actual value of the samples. As a result, the activation function and the selection of loss functions are strongly associated (Zhou et al., 2019). The proportionate penalty of a probabilistic false negative for a particular class can be changed by weighting the categorical cross-entropy loss function by class in multi-class

classification. These weights are determined by Class-Balanced Loss in proportion to the inverse of the number of samples in each class (Ho & Wookey, 2019). In this study, we implemented the categorical cross-entropy loss function, which is frequently implemented for multiclass classification problems. This loss function evaluates the level of alignment between the predicted probability distribution and the actual distribution by comparing the predicted class probabilities to the actual class identifiers (Murugan, 2018). The following is the formula for categorical cross-entropy, or CCE:

$$\text{CCE} = - \sum_{i=1}^N \sum_{j=1}^C y_{ij} \log(\hat{y}_{ij}) \quad (28)$$

where, N : Total number of samples, C : Total number of classes, y_{ij} : True label for class j of the i th sample (1 if the class is correct, otherwise 0), $\hat{y}_{ij}$: Predicted probability for class j of the i th sample, and $\log(\hat{y}_{ij})$: Logarithm of the predicted probability.

- Output Kernel Initializer:** The CNN layer weight initialization process is defined by the kernel initializer. In neural networks, weight initialization is crucial since it has a big influence on the training process's stability and rate of convergence. In this approach, the initial weights of the hidden layer i are sampled uniformly from a symmetric range that depends on the activation function used. For the *tanh* activation function, as demonstrated in Glorot and Bengio (2010), the interval is given by:

$$w_{ij} \in \text{Uniform} \left[-\sqrt{\frac{6}{fan_{in} + fan_{out}}}, \sqrt{\frac{6}{fan_{in} + fan_{out}}} \right] \quad (29)$$

where fan_{in} represents the number of units in the preceding layer, and fan_{out} represents the number of units in the subsequent layer.

- Optimizer:** Optimization refers to the process of minimizing or maximizing an objective function, also known as the cost or loss function. The primary aim is to identify the optimal parameters that effectively minimize the cost function over the training dataset of a deep neural network. To achieve this, gradient descent is often employed as a method for finding these optimal parameters (Du, Lee, Li, Wang, & Zhai, 2019). The optimizer's function affects a deep neural network model's accuracy (Desai, 2020). In this study we have used Adamax optimizer where the computation method of the second momentum part of ADAM is used in the AdaMax with its maximum value (Yi, Ahn, & Ji, 2020). This provides a much better opportunity. The AdaMax method is derived from the ADAM optimization algorithm and utilizes the maximum value for computing the second momentum. The equations governing AdaMax are as follows:

$$u_i = \max(\beta_2 \cdot u_{i-1}, \left| \frac{\partial C}{\partial w}(w_i) \right|) \quad (30)$$

$$w_{i+1} = w_i - \frac{\eta \hat{m}_i}{u_i} \quad (31)$$

where u_i represents the updated maximum value, β_2 is the exponential decay rate for the second moment, $\hat{m}_i$ is the bias-corrected first moment, and η is the learning rate.

- Learning Rate:** The learning rate is a crucial hyperparameter implies impact where the training dataset is too large so that the system cannot effectively generalize. This problem may also persist when there are too many neurons in the network such that while the computation requirements become much larger than the input dimensions in the vector space (Igiri, Anyama, & Silas, 2015). While it lessens the train loss it is actually a widely recognized fact that a deep network cannot be trained by a small

initial learning rate (Li, Wei, & Ma, 2019). The learning rate is important in relation to gradient descent which updates weight by the following equation:

$$W_{new} = W_{old} - \alpha \cdot \frac{\partial L}{\partial W} \quad (32)$$

where, W_{new} and W_{old} represent the updated and previous weights, respectively, α is the learning rate. and $\frac{\partial L}{\partial W}$ is the gradient of the loss function with respect to the weights. As for the coefficient for learning rate, the ‘ReduceLROnPlateau’ (Thakur et al., 2024) method is used in this study with the initial learning rate as 0.001. This means that the learning rate can be gradually adjusted within the model easily in order to enhance convergence rather than overfitting.

6. **Batch Size:** To develop a reliable convolutional neural network capable of producing the correct classification of images, the setting of multiple hyperparameters is required. The batch size meaning the number of images needed in order to do one complete forward and backward pass is one of the most important hyperparameters (Kandel & Castelli, 2020). Batches of higher size do not generally give high accuracy, the optimizer, and learning rate will also affect it. The network can easier train by lowering the learning rate as well as the batch size, especially at time of fine-tuning. In the present work, we have also chosen the batch size of 32 for training which strike a fine balance between intensity and effectiveness.
7. **Epoch:** The complete pass through the training dataset is called one epoch. In each epoch, the weights of the model are adjusted in the way that minimizes the loss over all training examples. Total time of training depends on the epoch numbers. If a model has a limited number of epochs, it may not pick up significant patterns from the data and, on the other hand, if it has too many epochs it picks the noise from the data. The number of epochs selected guarantees the model finds reasonably good solution and still serve the purpose of training efficiency (Brownlee, 2018; Srivastava, Hinton, Krizhevsky, Sutskever, & Salakhutdinov, 2014). In our study, we have utilized 100 epochs, which would guarantee that the employed model is properly trained and has achieved acceptable levels of convergence in relation to time taken in training, and convergence.

4.3. Evaluation metrics

1. **Accuracy:** A threshold-based metrics (Yacouby & Axman, 2020) which is the ratio of samples that have been accurately classified by the actuality or with precision to the total samples to be classified is called accuracy. As would be expected, this metric, which is also referred to as accuracy, is used also in multi-class issues in which there are more than a binary number of labels (Wang, Chu, & Xie, 2007). More preciously, The percentage of correctly classified predictions is measured by accuracy, which can be mathematically represented as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (33)$$

where, TP = True Positive: Correctly predicted positive instances, TN = True Negative: Correctly predicted negative instances, FP = False Positive: Incorrectly predicted positive instances and FN = False Negative: Incorrectly predicted negative instances.

2. **Precision:** The definition of precision, which quantifies the percentage of accurate positive predictions (Krasnodębska, Goch, Verstegen, Uhl, & Pesaresi, 2024), is as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (34)$$

It only indicates the number of relevant selected data components Only. That is, the ratio of the extent of true positive observations to the number of cases that positive was predicted by the algorithm (Hasnain et al., 2020).

3. **Recall:** It introduces “what number of relevant data items are selected”. More specifically, out of all the observations that are indeed positive, how many were assigned by the algorithm (Vakili, Ghamsari, & Rezaei, 2020). According to Eq. (35), the recall equals the number of true positives divided by the sum of true positives and false negatives:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (35)$$

4. **F1-score:** This measure uses the total number of true positives, false negatives, and false positives to account for all classes (Hinojosa Lee, Braet, & Springael, 2024)

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (36)$$

It is the harmonic mean between precision and recall, bringing a proper balance between them.

5. **Kappa:** Cohen’s Kappa coefficient (k), governing the accuracy of random classifier and by anticipated accuracy coefficient indicates the number of times two observers agree in the result on a given case and was designated in the machine learning model if the truth is grounded in data (Vujović et al., 2021). Comparatively, interpretation of Kappa ranges from -1 to $+1$ and the extent of the agreement caused by chance is also measured the same as AUC; a Kappa 0 being that of ‘Random Classifier’. It has been more popular within the context of machine learning because it can be applied to the cases of unbalanced datasets (Widmann & Roccato, 2020). It still has a strong relationship with higher values on balanced datasets, though, based on the recent work (Delgado & Tibau, 2019). In addition, it does not allow interpretation of prediction accuracy or comparability across various sampled datasets.

$$\text{fc} = \frac{(TN + FN)(TN + FP) + (FP + TP)(FN + TP)}{TP + TN + FN + FP} \quad (37)$$

$$\text{Kappa} = \frac{(TP + TN) - \text{fc}}{(TP + TN + FN + FP) - \text{fc}} \quad (38)$$

6. **Matthews correlation coefficient (MCC):** Another measure that is not sensitive to the problem of class imbalance is the Matthews correlation coefficient (MCC). This approach employs contingency matrix to estimate Pearson product moment (Chicco & Jurman, 2020). Expressed in terms of the entries of the confusion matrix, the MCC is calculated as follows:

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP) \cdot (TP + FN) \cdot (TN + FP) \cdot (TN + FN)}} \quad (39)$$

7. **Jaccard Index:** The Jaccard index of similarity of two sets is defined as the total number of elements that are in both the set (overlap in sets) to the total number of elements in both the set (Krasnodębska et al., 2024). In the classification case, it evaluates the proportion of correctly assigned positives relative to all positives in both the model and the reference:

$$\text{Jaccard} = \frac{TP}{TP + FP + FN} \quad (40)$$

8. **Confusion Matrix:** This matrix, one of the most comprehensible and transparent metrics available, can be used to determine the accuracy and correctness of an ML algorithm. Its main application is in classification problems where multiple types of classes may be generated (Sammur & Webb, 2011).
9. **AUC-ROC Curve:** This statistic indicates the discriminant capacity of the test and is based on the receiver operating characteristic curve (ROC curve), this curve represents the relationship established between the false positive rate (1 specificity) and the genuine positive rate also called sensitivity or recall. However, the ROC-AUC score can be used as a performance measure IF the values of the positive class and negative class are equal (Ho & Wookey, 2019).

Table 6
Train and test accuracy of different models.

Model		Train accuracy (%)	Test accuracy (%)
Attention based CNN		92.46	91.97
Classifier models	RF	100	99.04
	SVC	100	99.13
	LR	99.99	99.01
	KNN	99.69	99.35
	XGBoost	100	98.8
	MLP	100	99.24
ConvLSTM		96.97	94.25

5. Experimental outcomes

In our study, we used rice variety dataset to train the proposed model using a structure approach. The first approach therefore consisted of using an original attention system applied to a model that was able to combine linear and non-linear features from the data set. This guidance mechanism enabled the model to pay necessary attention to the significant parts of the images to classify rice as required. This was very important since the variations between the kinds of rice are very slight but very discriminative. Through successful weighting of important aspects while suppressing non-considered information, the attention mechanism helped the model to focus on certain parameters, thus giving shapes to intricate patterns that could not be easily detected.

Once the attention-based CNN model successfully extruded these critical features, we used them to train a series of baseline classifiers. These classifiers such as RF, SVC, LR, KNN, XG and MLP were selected based on the prior established success in performing supervised learning tasks. The feature extraction performed by the model proposed in this paper allowed for the high performance of these classifiers in assigning proper classifications to the rice varieties. By integrating this advanced feature extraction with relatively powerful classifiers, the improvements of our hybrid scheme were shown in this case.

The analysis of our results includes four key aspects: generalized analysis, classification analysis, statistical analysis and graphical analysis. These overall evaluations enabled us to conduct a multiaspect analysis of the model with regard to rice variety classification, providing an in-depth perspective on its performance and robustness.

5.1. Generalization analysis

This analysis assesses the generalization ability of models as implemented from training data to new test data. A model with good generalization is a model which a small difference or no difference at all between training set accuracy and test set accuracy is typical for a model that has not over-fitted and is actually learning meaningful patterns from the training set and not just the training labels. Table 6 shows the train and test accuracy of different models for the rice variety feature extraction and classification task, highlighting their generalization performance on the dataset.

The integration of an attention mechanism into a proposed hybrid model yielded a train accuracy of 92.46% and a test accuracy of 91.97% and had a very small generalization gap of only 0.49. This shows how it had a high ability of learning more complicated patterns in the data than has been used in training yet still had a high accuracy on the unknown samples. Due to the utilized attention mechanism, the model highlighted the key features that are important for the perception of data. This was possible alongside the hybrid architecture, making for a symmetrical learning process as well as minimizing the dependency problem and consequently diminishing overfitting. These results provide reception to the development of the proposed model and its applicability in handling the matters of rice variety classification accurately and efficiently.

Table 7
Classification metrics: Precision, Recall, and F1-Score for rice variety identification.

Model		Precision (%)	Recall (%)	F1-Score (%)
Attention based CNN		93.35	91.97	91.92
Classifier models	RF	99	99	99
	SVC	99	99	99
	LR	99	99	99
	KNN	99	99	99
	XGBoost	99	99	99
	MLP	99	99	99
ConvLSTM		94.34	94.25	94.26

The best set of models combining the lowest training time and highest testing accuracy is KNN and LR as they showed the least amount of variation between the two. KNN reached 99.69% training accuracy and 99.35% of test accuracy, the disparity here was only 0.34%. Likewise, for LR, the training accuracy was 99.99% and test accuracy was 99.01% with a difference of 0.98%. These minimal differences point to better stability and reliability that is characteristic of both models for the reliable classification of rice varieties.

Other models include Random Forest (RF), Support Vector Classifier (SVC), XGBoost and MultiLayer Perceptron (MLP) which all gave 100% of training accuracy which mean that these models learned perfectly the training set. But they showed small declines of the test accuracy indicating a mild overfitting. The largest gap belongs to XGBoost, which was built with a test accuracy of 98.8%, yet MLP shows better generalization among the models from this group with a test accuracy of 99.24%. These models, however, could be enhanced through other strategies such as parameter optimization and regularization to enhance the generation of results from unseen data.

Lastly, the ConvLSTM model generalization was the lowest where by training accuracy was 96.97% and the test accuracy was 94.25%, different by 2.72%. As such it tends to over fit and this calls for extension such as regularization, dropout or modification of architecture to increase its performance on unseen data.

5.2. Classification analysis

Rice varieties need to be classified correctly in order to improve agricultural productivity, distribution, and product quality. The classification analysis indicates the ability of the developed models to differentiate between twenty different types of Bangladeshi rices reflecting the assessment. Therefore, the choice of the metrics for evaluating a model's performance involves precision, recall and the F1-score in such classification problems. High precision reinforces the model by means of giving accurate indications; if it predicts a rice variety, it will probably be correct. This is particularly useful especially when classifying rice and much harm might occur if one variety is called the other including economic losses, inadequate processing or customer complaints. The recall, on the other hand, holds an ability to show how well the model can identify all the samples of certain type of rice, say, in the conditions of managing stocks or checking product quality. A good recall is likely to permit minimal cases of a given rice variety to go unnoticed. These metrics will give an overall insight of the general and specific varieties recognized by each models' strengths and weaknesses. Table 7 presents the classification metrics, including precision, recall, and F1-score, for various models used in rice variety identification, offering insights into their performance in accurately distinguishing between different rice varieties.

Since F1-score is the harmonic mean of precision and recall it gives a better insight as it excludes purely based on False Positives and False Negatives and shows how well the model can identify the rice variety and also minimize misclassifications. An F1-score on 1 means that the classifier works with the best balance of missing no true instances (high recall) while at the same time minimizing a high rate of wrong

identification (high precision).

For rice variety classification, RF, SVC, LR, KNN, XGBoost, and MLP were used which yielded high accuracy, and of 99% precision, recall and F1 score. The outcomes of these experiments indicate that it is feasible to achieve high classification accuracy using most of the traditional machine learning classifier models in designing rice varieties. However, the above results have shown that all the models under investigation have achieved a nearly perfect performance on both training and test datasets. While this is positive insight in the absence of new unseen rice varieties on which the models will be applied, such high performance might be suggesting that these models are overfitting. This is particularly possible, if the model was trained with more focus on the overfitting of the existing data which makes the model to have a hard time when called upon to work on the new data which has not been put through the training model. Thus, although these models can do well with recognizing well-known varieties of rice, they may not do as well if faced with new data or data subjects that are more diverse.

Although the performance of the Attention based CNN model with the attention mechanism was marginally lower than baseline classifiers, it was essentially generalizable for different varieties of rice. The Attention based CNN model achieves an F1-score of 91.92% for the client data, and the small difference between the training and the testing accuracy proves that the model minimizes the trade-off between overfitting and underfitting. It is because of that the attention mechanism helped our model in attending to the pertinent features, which is useful in differentiating between rice varieties, but not be sidetracked by noise or other unimportant information. This makes the model a good candidate for practical utilization in real life because the rice varieties that may be available might be slightly different from what was used while preparing the model data due to issues to do with environment, processing or some other factors that may have not been foreseen.

A deep model which utilizes novel structure called ConvLSTM, a convolutional layer to extract features from images and LSTM to model the temporal order of data gave adequate classification with F1-score of 94.26%. This model showed slightly lower precision and recall compared to previous models which can be promising when temporal data is present the model can show improvement for instance in the assessment of rice quality over time or multi-spectral data.

5.3. Statistical analysis

Statistical analysis is vital for selection, particularly in rice varieties, as such, in this study, we compared different models using Kappa, MCC as well as Jaccard Score/Index to gain insights into their efficiency in classifying rice varieties. These statistical metrics depart from the usual approach to accuracy by providing a representation of the overall classification performance indicating the number of hits and misses, taking into consideration the presence of imbalanced classes or simply sample variation. The Kappa coefficient compares the predicted and actual labels and gives a clue to the stability of the model's performance taking into account random or chance agreement. The MCC improves on classification accuracy measurements by using all four parameters of the confusion matrix; true positive, true negatives, false positives, and false negatives. Last of all, the application of the Jaccard Index checks the parameter of how successfully the models can identify the right classes out of the available classes. Table 8 implies the statistical metrics utilized in models.

The quantifiable measure that used to compare the proportions of actual and predicted classification profiles while accounting for the level of chance was the Kappa coefficient with the Attention based CNN applications of Kappa of 91.55%. This shows high accuracy of the model when predicting the actual labels of the new data set which means that in general it performs very well. The MCC score of 91.63% for the Attention based CNN model also adds more strength to the work as MCC is more favorable than other measures as it cares positive

Table 8

Kappa, MCC, and Jaccard index for different models in rice variety classification.

Model	Kappa (%)	MCC (%)	Jaccard Index (%)
Attention based CNN	91.55	91.63	86.44
Classifier model	RF	98.99	98.99
	SVC	99.08	99.08
	LR	98.96	98.96
	KNN	99.31	99.31
	XGBoost	98.74	98.74
	MLP	99.20	99.20
ConvLSTM	93.68	93.68	89.13

as well as negative predictions in the confusion matrix. The Jaccard Index of 86.44% inform that the predicted model successfully identified the positive class instances with high accuracy although the scores are slightly lower than those scores of conventional models, it suggests that there is a scope for enhancement of identifying the positive instances.

For the baseline classifiers with RF, SVC, LR, KNN, XGBoost, MLP, Kappa, MCC and the Jaccard Index showed high value with a percentage range, including 98.74%, 99.31% for Kappa, MCC and 97.67%, 99% for the Jaccard Index. The values expressed by these models assert that the models are very accurate in providing the right results in terms of true positives or true negatives, with minimum cases of false positive or false negatives respectively. However, such perfect performance is a bad sign that may suggest overfitting especially seeing that these models appear to learn the training data almost to perfection to maybe the total detriment of their ability to generalize on unseen data.

On the other hand, for the ConvLSTM model aimed at capturing temporal features, the Kappa was 93.68%, MCC 93.68%, and Jaccard Index of 89.13%. Despite these numbers, which represent still reasonable scores, this experiment shows that the ConvLSTM model is less successful compared to the conventional classifiers in terms of classification efficiency and correlation with true positive results. This indicates that although ConvLSTM can be useful for sequential or time series data, which is the case in this study, they may need a little more work in order to deliver the best results as the other models in this rice variety classification problem.

5.4. Graphical evaluation

The multi-faceted assessment of the Attention-Based hybrid model for rice classification was conducted in the following ways. The Confusion Matrix proved good recognition rate with few confusing classes in appreciating twenty different rice types. This strong performance was complemented by the computation of the ROC curves which offered important information about the differentiation capability of the models used when it came to different rice classes. Moreover, plots of learning curves which show the changes of accuracy and loss during training provided an idea of the training of current model and possibility of further optimization. Results in this research are represented not only by figures and mathematical equations connoting model behavior, but also by graphics and curves reflective of the proposed model's performance, which makes the analysis comprehensive. Apart from this, the findings of this research would help in developing further strategies to enhance the accuracy and stability of the rice classification model in the future, as well as shed light on the determinants of the effectiveness of the described approach.

The accuracy curve from Fig. 6 gives the insights by giving a straight view of the model in classification aptitudes. Training accuracy increases steeply and approaches 100% at epoch 40, indicating that the current model possesses a high capacity to detect the characterizing traits of the rice varieties in a training dataset. Validation accuracy becomes constant at 90%–95% illustrating good generalization on unseen data. Nonetheless, slight variations in the validation accuracy indicate that there may be problems with margin or rare samples

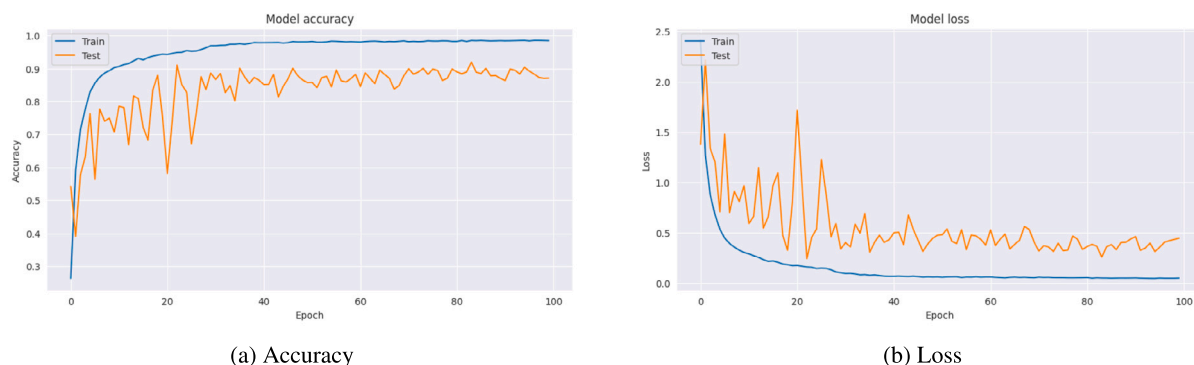


Fig. 6. Accuracy and loss for proposed attention-based model.

sometimes. These variations could be due to high within group variation like variations in lighting or alignment of grains or similarity between classes. Despite the small difference between the training and validation accuracies it is clear there is potential to improve the way of dealing with some of the difficult samples in the validation set. To fill the gap, solutions which might be applied are curriculum learning, where the model works with gradually increasing complexity of input examples, or ensemble methods, where predictions from several models are combined.

The loss curve from Fig. 6 describes the learning of the model during the process of training of the model. First, the overall cumulative loss falls steeply mainly because the model learns relatively simple and informative features and relationships of the training set in the initial training phase. The convolutional layers for spatial feature learning and attention modules for paying attention towards important features of the channel as well as the particular edge zone also play to this end in enhancing this progress. The test loss also has a similar pattern but is characterized by occasional sharp fluctuations, indicating a situation when the model compares unfavorably with certain validation samples. Such fluctuations might be in part due to noisy/low signal-to-noise ratios of the data in the test set, or the presence of classes which were not sufficiently represented in the training set. Nevertheless, the loss increases and finally converges and forms a tight band with the training and validation datasets, and there is little overfitting. However, the two or three jumps in validation loss prove that it is necessary to stabilize the model through data augmentation, class balancing or even fine-tuning of the model in view of misused examples.

Based on the assessment that we are going to conduct in this research; KNN is clearly the best performing classifier for rice variety identification. As we can see from Table 6, its generalization ability is very strong, the training accuracy can reach 99.69%, the testing accuracy as 99.35%, and the generalization gap is only 0.34, the smallest. Such consistency helps to emphasize the high generalizing ability of KNN, thus, opportunities for its failure when working with new data sets are slim. Such stability is highly desirable in such agricultural uses since accurate identification of rice varieties has a direct bearing on crop growing, distribution, and business forecasts.

Analyzing Fig. 7 of the ROC curve more deeply and intensively show much more astonishing evidence in the case of KNN. Completely message-naïve KNN kept its extraordinary ability to classify all data. There is no over-aging of rice samples under the microstructure developed. Thus, the model indicates perfect discriminatory ability, with an Area Under the Curve (AUC) of 1.00 for all the 20 rice varieties. Every ROC curve lies close to the upper-left quadrant, perfect indication of high sensitivity (true positive rate) and specificity (true negative rate). This result validates that KNN outperforms any other classification algorithm in one of the essential steps, namely the ability to differentiate feature differences between rice varieties that may be visually indistinguishable from each other, especially when different varieties of rice co-exist.

The confusion matrix shown in Fig. 8 of classifier KNN provides the general assessment of effectiveness of the classifier and the regions of its mistake making in the task of rice variety classification. The classifier tested perfectly out of the 6750 samples available for analysis, classifying 6706 of them accurately, and misclassifying only 44 of them, hence the accuracy being an astonishing 99.35%. However, the misclassified occasions are not absolutely equal, increasing appreciation of complex connections between some rice types which share similarities.

When analyzing the cases of misclassifications, several patterns rise from the analysis. From Table 3 out of 337 instances Shampakatari, for instance, misclassified 5 instances as Katarivog with this variety and 1 with Katari_Polao. In the same way, out of 338 samples Ganjiya misclassified 1 sample as Shorna5, which implies a slight uncertainty. The more extensive misclassifications occurred in the case of the BR varieties which possess some morphological and architectural similarities. In the same manner, out of 337 samples 7 were misclassified by BR28 with BR29, 1 as Katarivog and, 1 as Jirashail while, out of 338 samples 14 instances were misclassified by BR29 with BR28 as well as 1 as Katarivog, Gutisharna and 1 as Jirashail. These trends indicate that the closer relatedness of the varieties poses higher risks of mis-classification though the model performs impressively overall.

Pajam showed confusion with BR29, out of 338 samples it misclassified 2 instances, perhaps because the two are morphologically similar to the BR varieties. Lal_Aush out of 338 samples misidentified one example as Katarivog and one as Gutisharna whereas Gutisharna out of 337 samples misidentified two examples as Katarivog, one as BR28 and one as Lal_Aush. These patterns also underlined the fact that it is rather difficult to clearly relate some of these varieties to each other based on the extracted characteristics. Finally, Amon out of 337 samples misclassified 1 instance as Lal_Binni, a pair which may as well may belong to the same category of similar instances.

A notable feature of KNN classifier's results is the proximate performances: the classifier was able to classify all test samples of the 11 out of the 20 rice varieties in the datasets, and misclassified no samples of these classes. These varieties are Subol_Lota, Bashmoti, Shorna5, Chinigura_Polao, Katari_Polao, Lal_Biroi, Red_Cargo, Nazir-sail, Katari_Vog, Bashful, Lal_Binni. That the perfect classification was achieved for these classes suggest that the model was able to sort these classes well by identifying their distinct pattern hence affirming its ability to distinguish other patterns as well. This result also indicates that the classifier is good at categorizing some varieties of rice without ambiguity, most probably because the features used in the training process are distinct for that variety. These kinds of results also go further to justify the applicability of the hybrid feature extraction technique in the actualization of the framework in resourceful circumstances especially when the margins between classes are well defined.

These features are extracted from the proposed attention-based model, and KNN makes it particularly effective. The practical features of this algorithm are rich and nonlinear, which makes it possible to depict the exact and intricate features of the data patterns accurately.

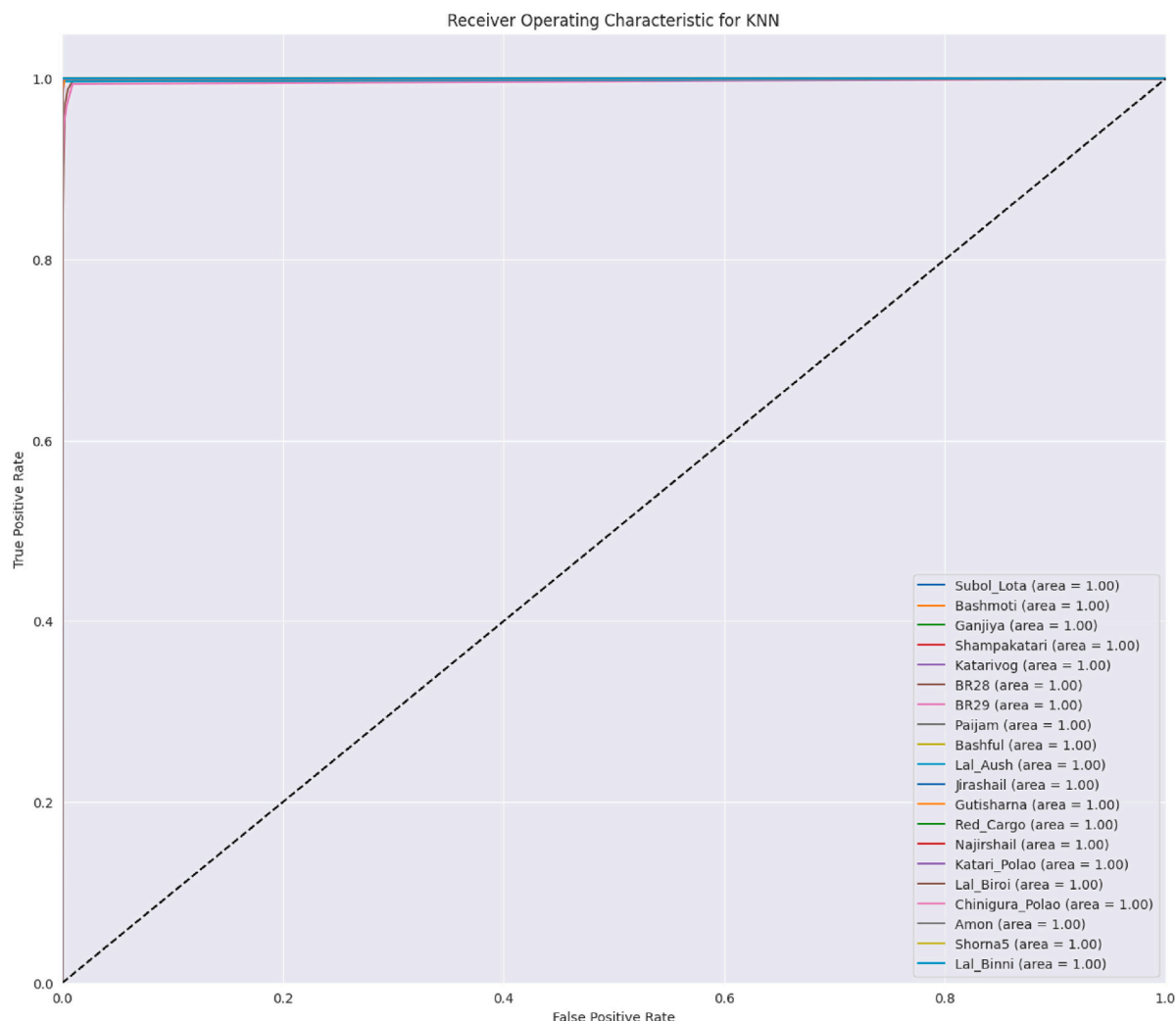


Fig. 7. AUC-ROC curve for the KNN classifier.

Higher Jaccard Index, Kappa score and MCC have also been reported in previous we evaluations that reaffirm that KNN has aptitude to perform the complex chore of rice variety classification with high reliability.

In generalizing the collective results and comparing the models together shown in Fig. 9, we get a clear picture out of the obtained accuracy, precision, recall, and F1-score. The KNN model can clearly be seen as being quite unique from all of the other models presented, it secured a test accuracy of 99.35% and all the metrics were exceptional throughout, securing a precision of 99%, recall of 99%, and an F1-score of 99. These metrics support the fact that generalization ability of this model is high and confirms that proposed classifier is suitable for rice variety identification. Other conventional models such as SVC, RF, and MLP also perform remarkable, all their classification metrics standing at 99% level which show the effective training from the dataset, though marginally overtrain the model. However, the proposed hybrid model, does well to establish reasonable accuracy to the system and at the same time Generalize well by realizing a 93.35% accuracy, 91.97% recall, and a 91.92% f1 score. ConvLSTM has presented a lower generalization gap and it achieved almost to the level of the best model with an F1-score of 94.26%.

From a statistical analysis point of view, differentiating the model's Kappa, MCC, and Jaccard Index also gives the researcher further understanding of the degree of parity the models have in classification shown in Fig. 10. KNN is again at the top position with the best values

of 99.31% for Kappa and MCC and a Jaccard Index of 98.73% for all data splits. SVC and MLP are close behind, and both models achieve a Kappa as well as MCC over 98.9%, which underlines the high efficiency of the models. The proposed hybrid model is competitive, and yielded high Kappa of 91.55% MCC of 91.63% and low Jaccard of 86.44% showing generalizability and classification accuracy. ConvLSTM also seems to perform relatively well but shows a slight lower Jaccard Index of 0.8913 and reveals the that aspect of temporal modeling still requires further optimization. Combined with such visualized results, these findings underscore the combined strengths of KNN as the most accurate model and the hybrid solution's ability to provide scalable and generalized outcomes. When compared in this manner, it becomes very easy to see relative trade-off between qualitative classifiers and general ones in terms of accuracy and usability.

6. Conclusion

The Attention-Based Hybrid Model proposed in this research contributes a major leap in the classification of Bangladeshi rice varieties. The problem of distinguishing morphologically similar rice varieties with high accuracy and reliability is solved due to the integration of a Convolutional Neural Network (CNN) and a Convolutional Block Attention Module (CBAM). The attention mechanisms prevent the model from considering all the spatial coordinates and channel dimensions at

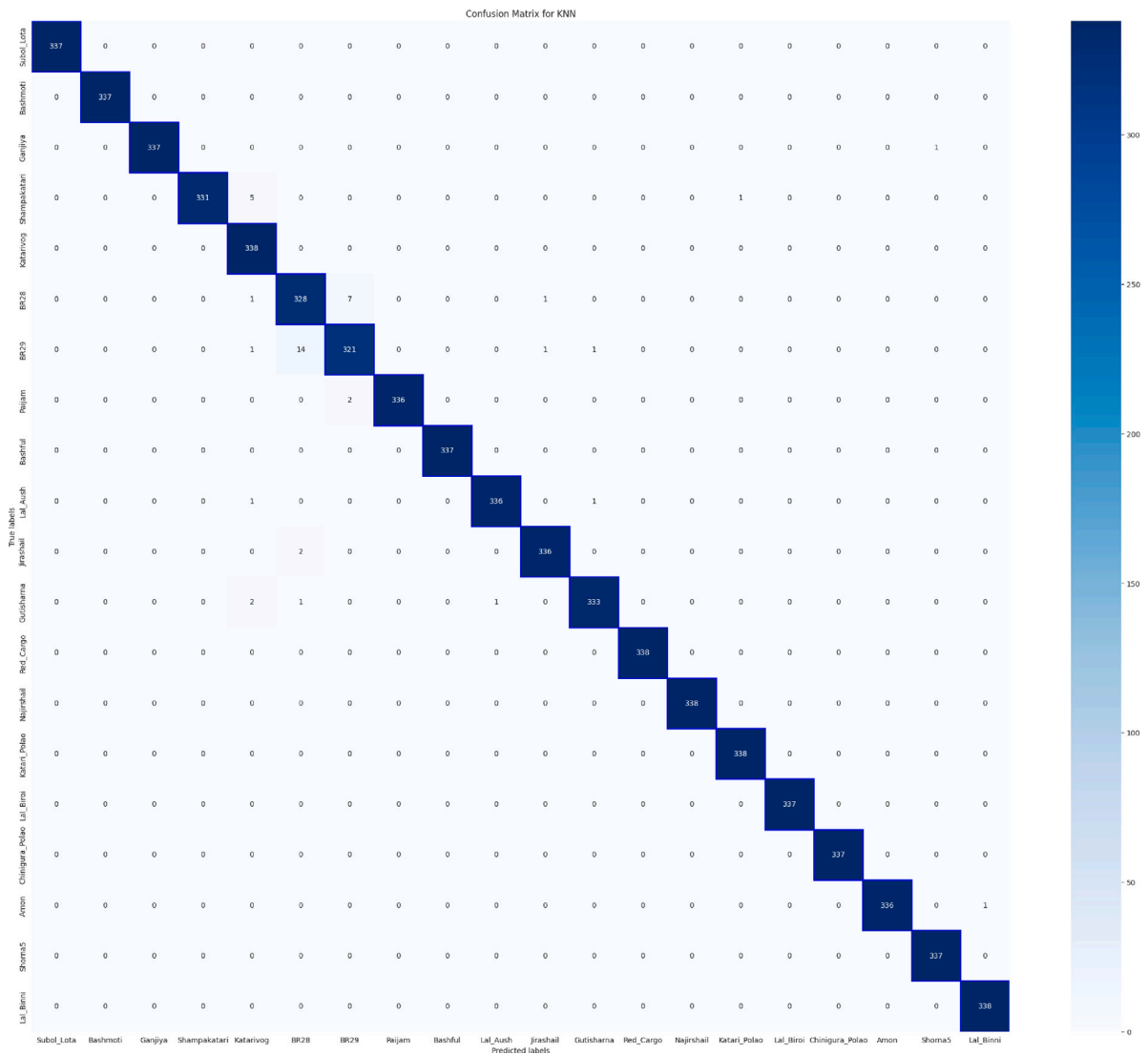


Fig. 8. Confusion metrics for the KNN classifier.

once but directs the model to focus on the most relevant map areas and channels, which in turn yields to accurate feature extraction and high classification accomplishments. The Accuracy of the Attention Based CNN was 91.92% for F1-score and in the same way, the KNN classifier gave exceptional accuracy of 99.35% thus emphasizing on the combination of effective feature extraction as well as classification. This hybrid approach surpasses conventional approaches in terms of feature discrimination insufficiency and inability to scale adequately. This work engages a versatile, effective and reliable system for automating agricultural processes and showing the possibilities to improve product yields, their quality, and sales appeal in the sphere of agricultural produce. In doing so, this research establishes solid footing towards the general adoption of deep learning and attention mechanisms to even complex applications in automated systems and precision agriculture.

Future investigations can aim at incorporating real time function which will classify and assess the quality of rice grains as they are produced in farming sector and used or processed in industries. There is an option to perform multispectral or hyperspectral imaging to enrich spectral data in order to help the model better differentiate closely related types. In addition to the improvements from a larger

sample of rice, more diverse rice types and environmental conditions could also improve generalization and practical applicability to other agricultural environments. In its current form the model could be deployed on resource-starved devices such as smartphones or edge devices, allowing users like farmers in remote areas to access it. Further, applying the proposed framework to other agricultural problems, a pest identification, yield estimation, and quality sorting for various crops, for instance, would show flexibility of the proposed framework. By using explainable AI techniques, automated agricultural systems may become more transparent and trustworthy by improving the model's interpretability. By following these paths, the suggested model can develop into a thorough, flexible instrument that promotes sustainable farming methods and meets the rising need for food security and quality around the world.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

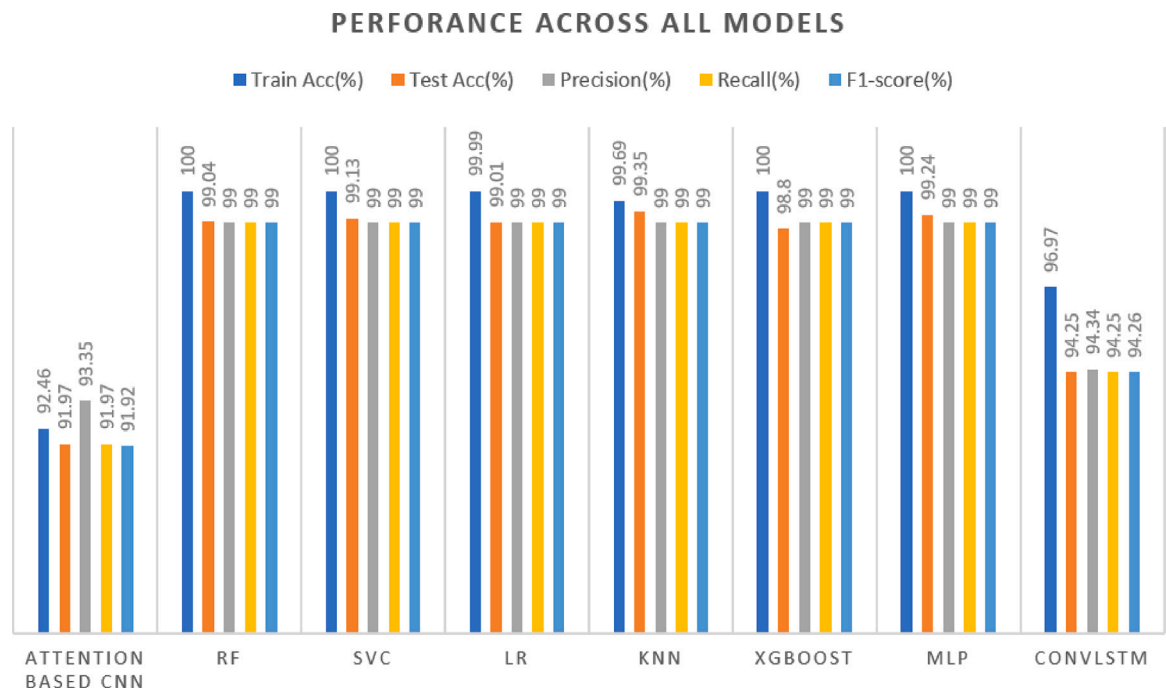


Fig. 9. Performance measures for all models.

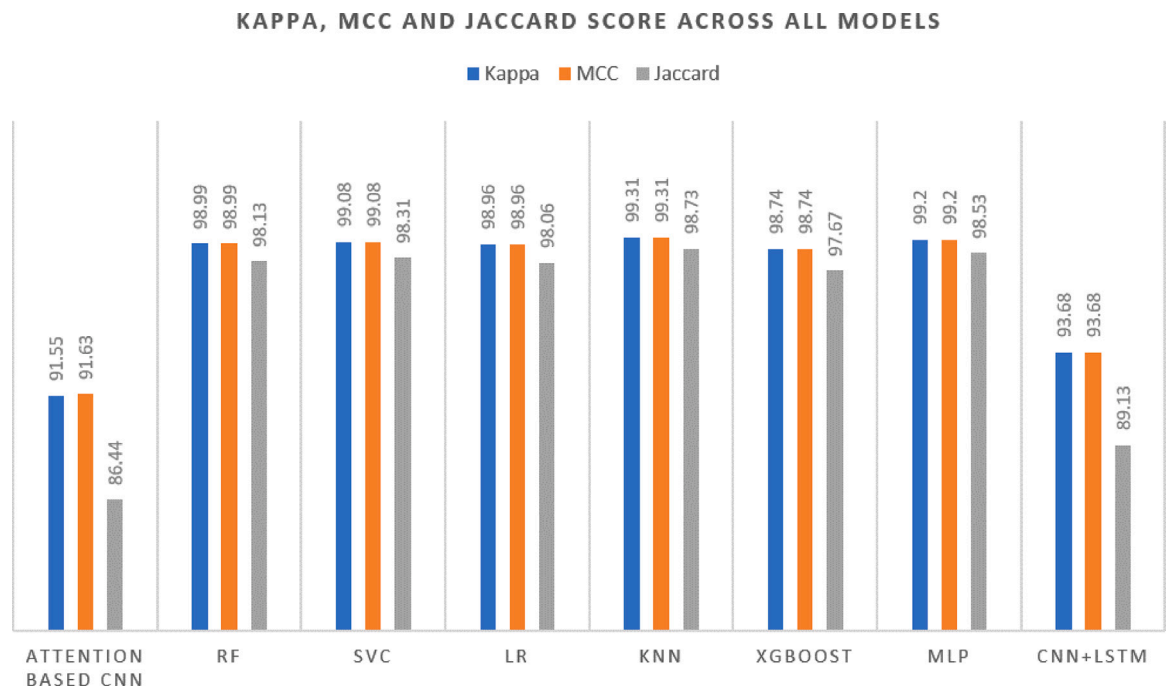


Fig. 10. Kappa, MCC and Jaccard score for all models.

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